CAITG WINTER SCHOOL

Coding text with GenAI — hands-on

The new Annotation tool in LDaCA Wordflow, a code-free text analytics app

Dr Chao Sun · Sydney Informatics Hub

Thursday 30 July 2026 · 45 minutes, mostly at your keyboard

START THIS NOW — `SIH.TOOLS/WORDFLOW`

Desktop app — v0.7.1

Mac & Windows installers on the page. Everything runs on your own machine — **recommended**, and the only option if you don't have an Australian university login.

Binder (browser)

Nothing to install, runs in the cloud. Needs an **AAF sign-in** (Australian university credentials) — not available to international participants. Don't upload sensitive data.

Get one of these **launching now** — it can install or spin up while I talk for the next few minutes.

(There's also `pip install ldaca-wordflow` for Python users — same app, no advantage if you don't already live in Python.)

WHO BUILT THIS?

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All at the **University of Sydney**, a **partner institution** of the **Language Data Commons of Australia (LDaCA)** — a **co-investment partnership with the Australian Research Data Commons (ARDC)** through the HASS and Indigenous Research Data Commons. The ARDC is enabled by the Australian Government's **National Collaborative Research Infrastructure Strategy (NCRIS)**.

WHY BUILD WORDFLOW?

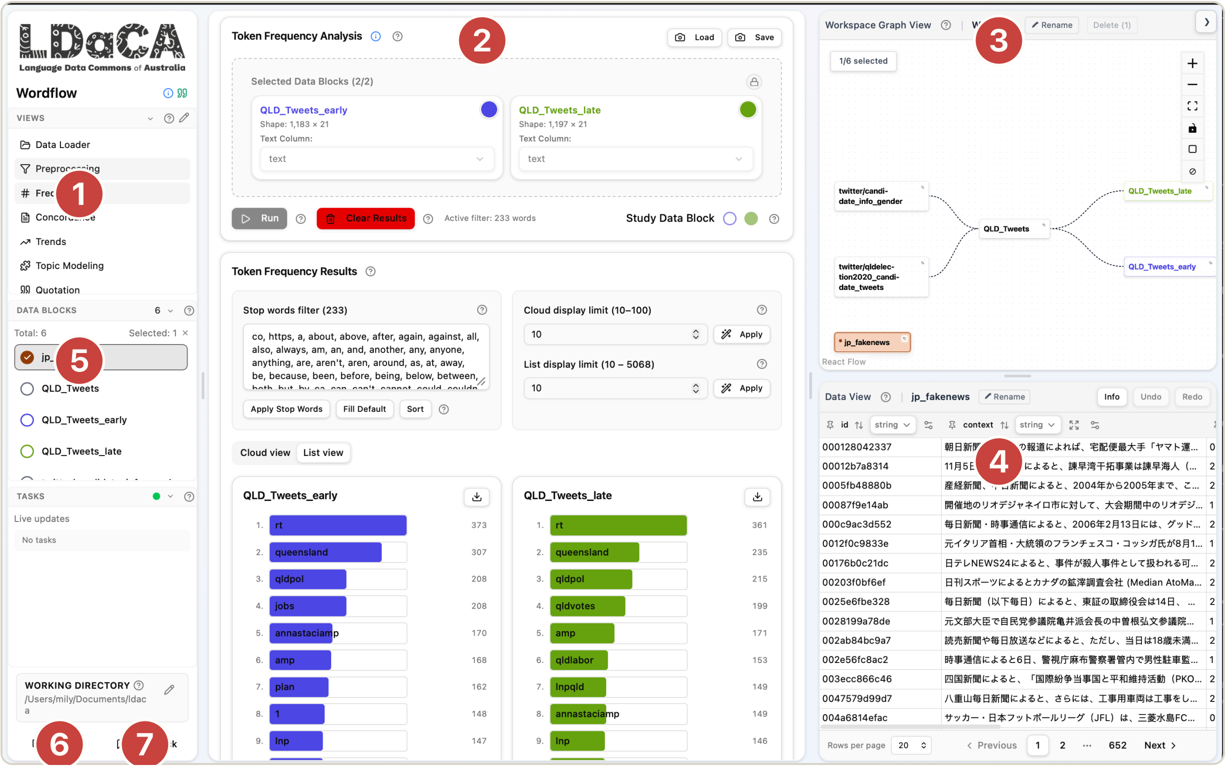
Modern text analysis — corpus methods, NLP, and now GenAI coding — mostly lives in **Python and R**. That's a wall for most researchers who work with text.

Wordflow puts those methods behind **point-and-click tools**: your data stays visible, every step is inspectable, and no code is required.

> Text data flows through stackable, single-purpose tools — the meaning of an analysis is shaped by how you shape your data, not by the tool itself.

Today's tool — **Annotation**, new in v0.7 — applies what you heard in the first half of this session: prompting, previewing, intercoder reliability. Without writing API code.

THE INTERFACE — 7 THINGS TO KNOW



1. **Tool sidebar** — pick a tool here
2. **Tool interface** — controls and results for the selected tool
3. **Workspace graph view** — where your current project's data and its analytical derivations live
4. **Data viewer** — manage and inspect columns and rows in a data block. Not just a viewer.
5. **Data block quick-select** — switch the active block without going back to the graph
6. **Tutorial** — opens the tutorial index page; or click the ⓘ icon next to any control to jump straight to its section
7. **Feedback** — tap anytime something is confusing, surprising, or broken

ONE IDEA TO HOLD ONTO

Everything is a data block.

- Your imported dataset is a data block. A filter, a join, an analysis result — each becomes a **new block** in the workspace graph.
- Every tool is a **lens**: it reads a block, and its results can go back into the workspace to feed the next tool.
- The **Annotation tool** plays the same game — it adds **annotation columns** to a block, so the AI's coding is just more data you can filter, compare, and analyse.

That's the deck done — the rest of the session is live in Wordflow.

THE HANDS-ON, IN ONE SLIDE

1. **Create a workspace**, import the sample data — the **QLD Election 2020 Twitter** blocks. (Or drop in your own CSV / text files. Excel & zip import is broken in 0.7.1 — CSV works.)
2. **Annotation tool** → candidate info block → **Start new annotation** → column `gender.ai`.
3. **Codebook → Create New → Edit** — define the codes: `M` / `F` / `U`, each with a description. Try a few rows by hand in **Manual** mode first.
4. **Flip to AI** — Add Provider (e.g. **OpenRouter**) with **your own API key**, then pick a model suited to your coding task. It doesn't have to be Gemini — those were just the low-cost models enabled for the live session.
5. **Preview → Compare To → confusion matrix** — check agreement with the human coding, find the weak spots, revise your prompt.
6. **Run All**, review and correct. Then: **your own data, your own coding problem.**

TAKE IT HOME

- **The shared workshop API key has been removed.** For your research: use **your own API key** (OpenRouter, or a provider your institution approves) — or a **local model**, so your data never leaves your machine.
- **Check your ethics approval before you code real data with AI.** Which models and providers are permitted — and whether your data may be sent to an external API at all — is an ethics question, not a technical one.
- **Everything you learned in the first half applies here** — pilot on a sample, measure agreement against human coding, revise the codebook, document your prompts.
- **Keep Wordflow** — `sih.tools/wordflow`. Tap the in-app **Feedback** button freely; cite via the quote icon in the sidebar ("Cite LDaCA Wordflow").

Data: Bruns, A.; Angus, D.; Cohen, T.; QUT Digital Observatory (2022). *Queensland Election 2020 on Twitter.* QUT. doi.org/10.25912/RDF_1665115527020 · Gender metadata: Sydney Corpus Lab.